KAIZEN MATHS

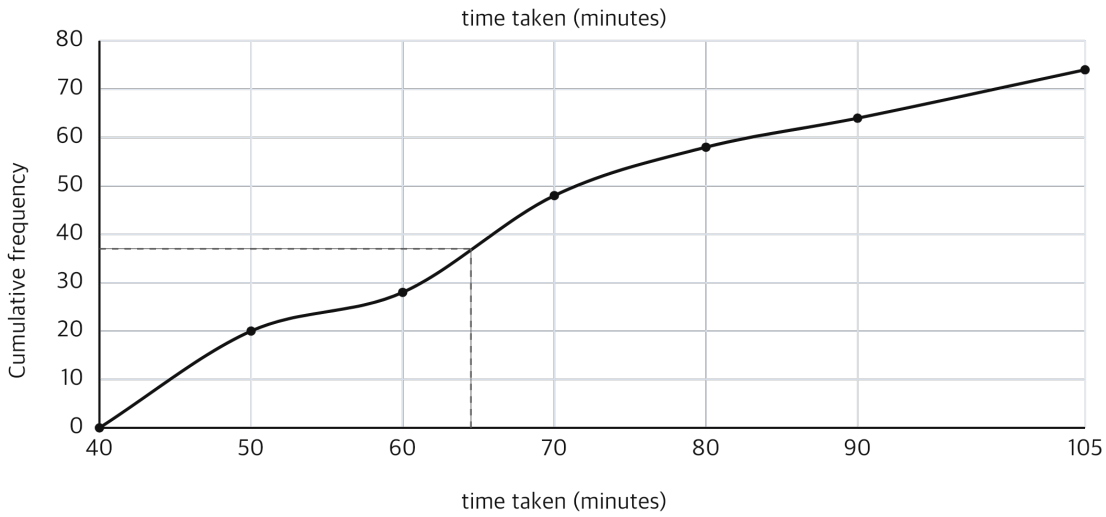
Lesson starter

Cumulative Frequency Curves and Ogives

Reading Curves

Name _____

SA1



Estimate the median.

STA-5424-L2 · SA

Show your working

KAIZEN MATHS

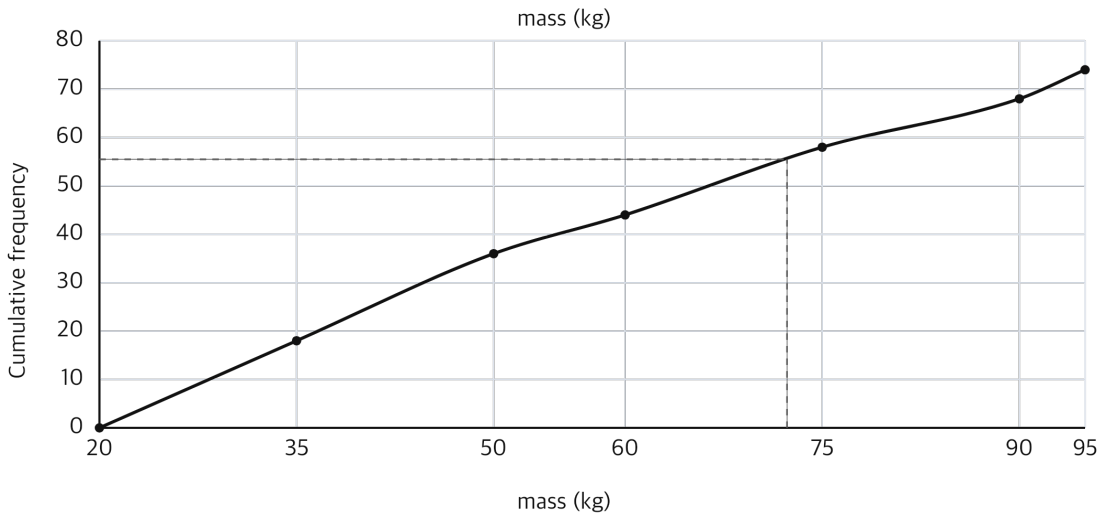
Lesson starter

Cumulative Frequency Curves and Ogives

Reading Curves

Name _____

SB1



Estimate the upper quartile.

STA-5424-L2 · SB

Show your working

KAIZEN MATHS

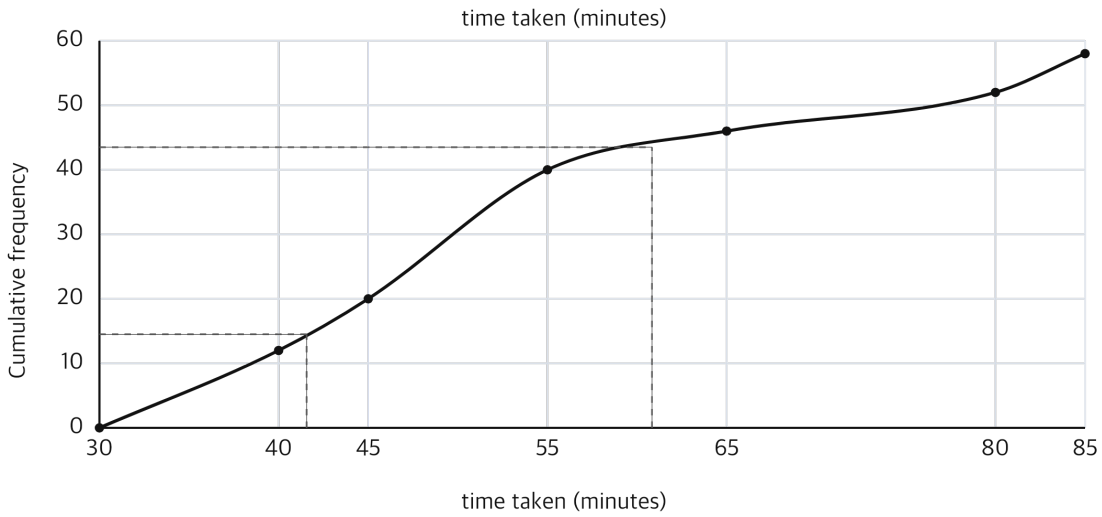
Lesson starter

Cumulative Frequency Curves and Ogives

Reading Curves

Name _____

SC1



Estimate the interquartile range.

STA-5424-L2 · SC

Show your working

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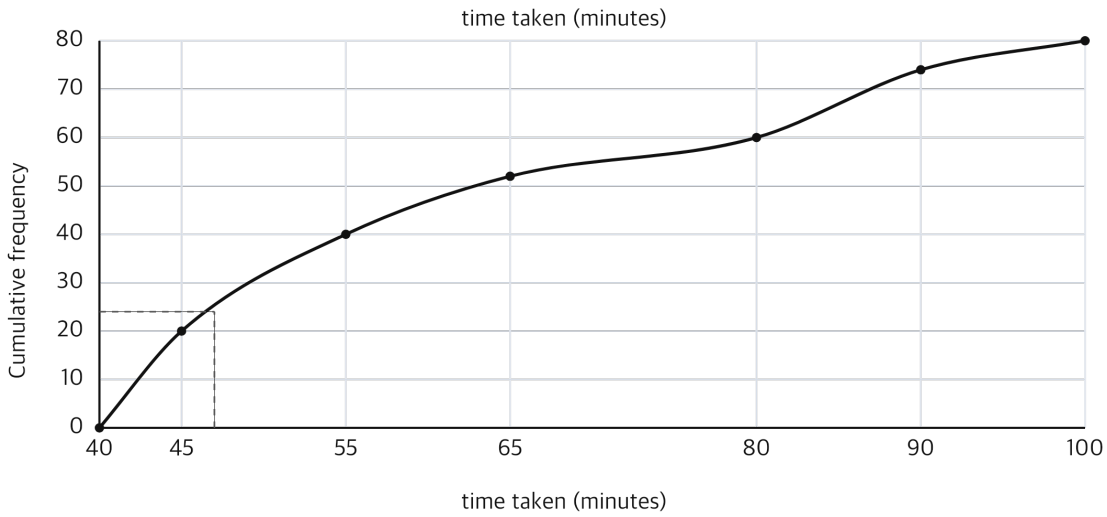
Lesson starter

Cumulative Frequency Curves and Ogives

Reading Curves

Name _____

SD1



Estimate the 30th percentile.

STA-5424-L2 · SD

Show your working

KAIZEN MATHS

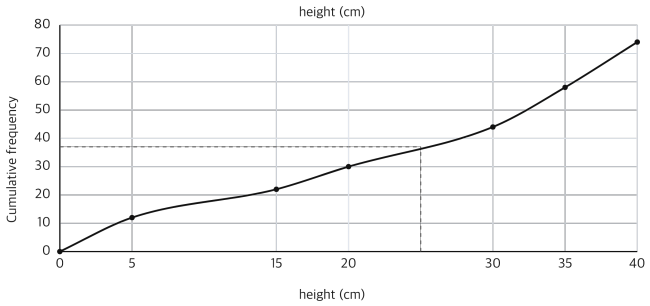
Exit ticket

Cumulative Frequency Curves and Ogives

Reading Curves

Name _____

EA1



Estimate the number of plants with height below 25 cm.

STA-5424-L2 · EA

Show your working

KAIZEN MATHS

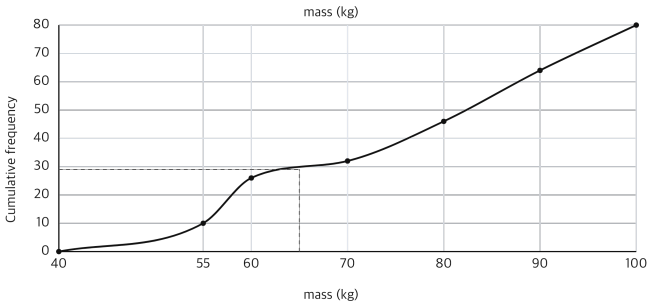
Exit ticket

Cumulative Frequency Curves and Ogives

Reading Curves

Name _____

EB1



Estimate the number of parcels with mass above 65 kg.

STA-5424-L2 · EB

Show your working

KAIZEN MATHS

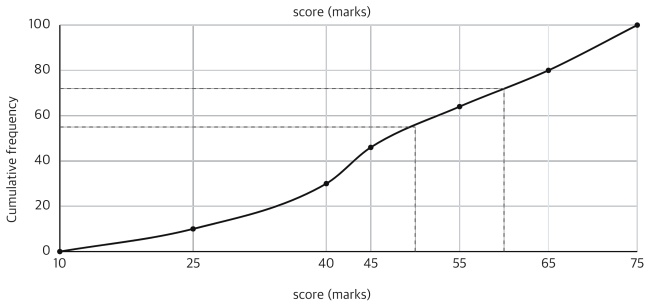
Exit ticket

Cumulative Frequency Curves and Ogives

Reading Curves

Name _____

EC1



Estimate the number of candidates with score between 50 and 60 marks.

STA-5424-L2 · EC

Show your working

KAIZEN MATHS

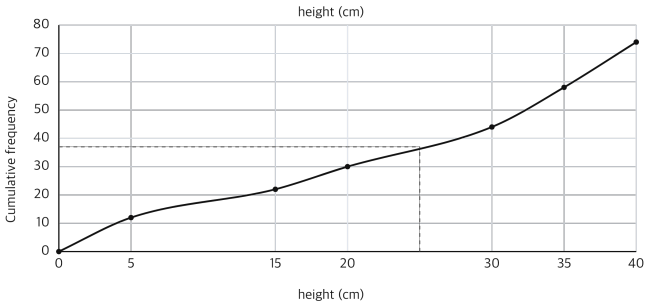
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Cumulative Frequency Curves and Ogives

Reading Curves

Name _____

EA1



Estimate the number of plants with height below 25 cm.

STA-5424-L2 · EA

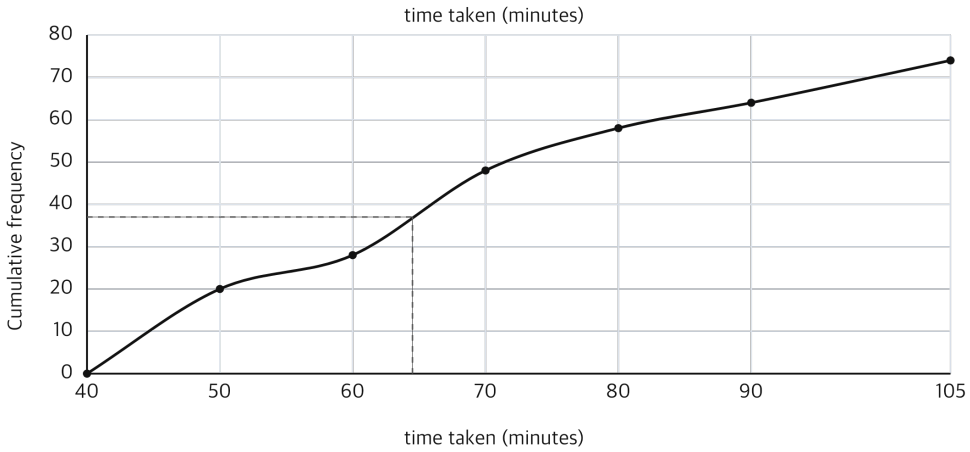
Show your working

Teacher answers & working

Cumulative Frequency Curves and Ogives · Reading Curves

STA-5424-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Estimate the median.

Answer:

64.5 minutes

Working:

The total frequency is 74.

Find the cumulative frequency position for the median.

median position = 37

On the curve, go across from this cumulative frequency to the curve, then down to the horizontal axis.

median ≈ 64.5 minutes

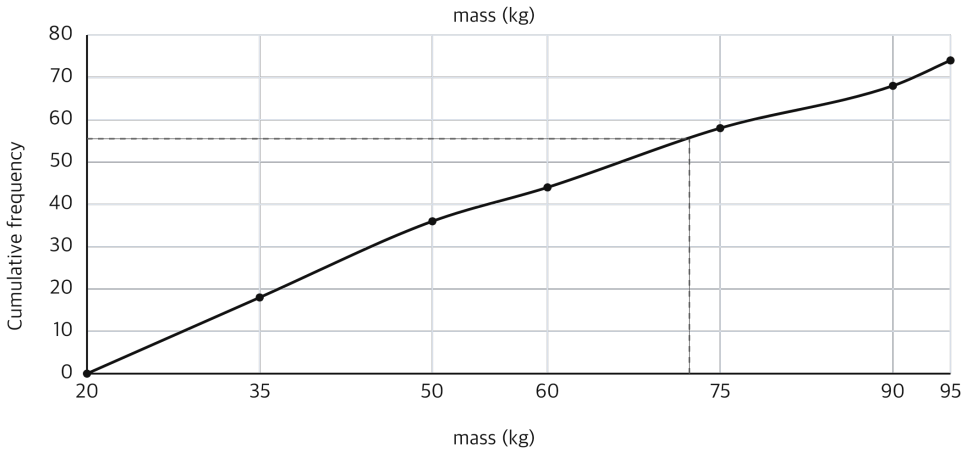
64.5 minutes

Teacher answers & working

Cumulative Frequency Curves and Ogives · Reading Curves

STA-5424-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Estimate the upper quartile.

Answer:

72.3 kg

Working:

The total frequency is 74.

Find the cumulative frequency position for the upper quartile.

upper quartile position = 55.5

On the curve, go across from this cumulative frequency to the curve, then down to the horizontal axis.

upper quartile ≈ 72.3 kg

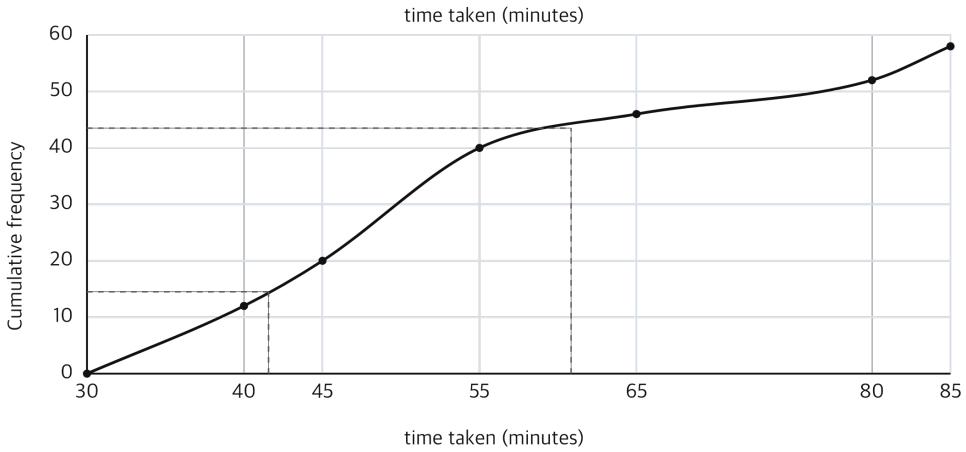
72.3 kg

Teacher answers & working

Cumulative Frequency Curves and Ogives · Reading Curves

STA-5424-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Estimate the interquartile range.

Answer:

19.3 minutes

Working:

The total frequency is 58.

Q1 position = 58 ÷ 4 = 14.5

Q3 position = 3 × 58 ÷ 4 = 43.5

Read Q1 and Q3 from the cumulative frequency curve.

Q1 ≈ 41.6 minutes

Q3 ≈ 60.8 minutes

IQR ≈ 60.8 - 41.6 = 19.3 minutes

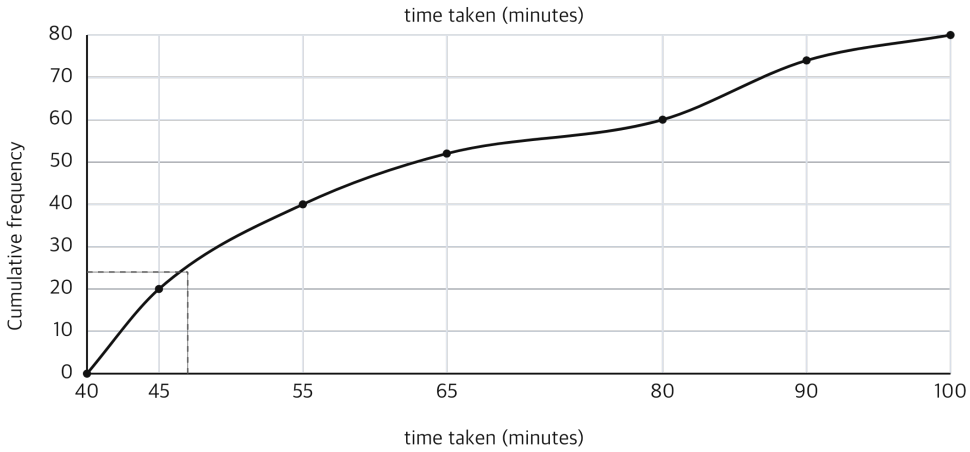
19.3 minutes

Teacher answers & working

Cumulative Frequency Curves and Ogives · Reading Curves

STA-5424-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Estimate the 30th percentile.

Answer:

47 minutes

Working:

The total frequency is 80.

Find the cumulative frequency position for the 30th percentile.

30th percentile position = 24

On the curve, go across from this cumulative frequency to the curve, then down to the horizontal axis.

30th percentile ≈ 47 minutes

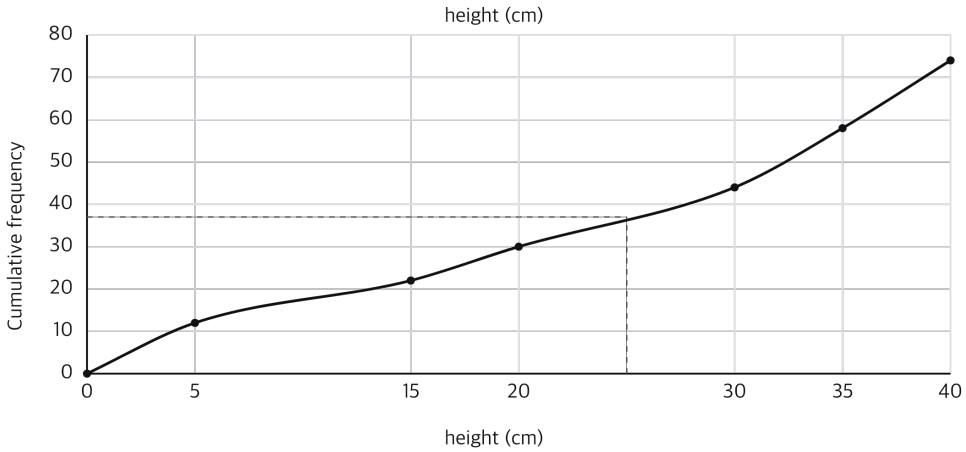
47 minutes

Teacher answers & working

Cumulative Frequency Curves and Ogives · Reading Curves

STA-5424-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Estimate the number of plants with height below 25 cm.

Answer:

37 plants

Working:

Read from 25 cm up to the curve, then across to the cumulative frequency axis.

number below 25 ≈ 37

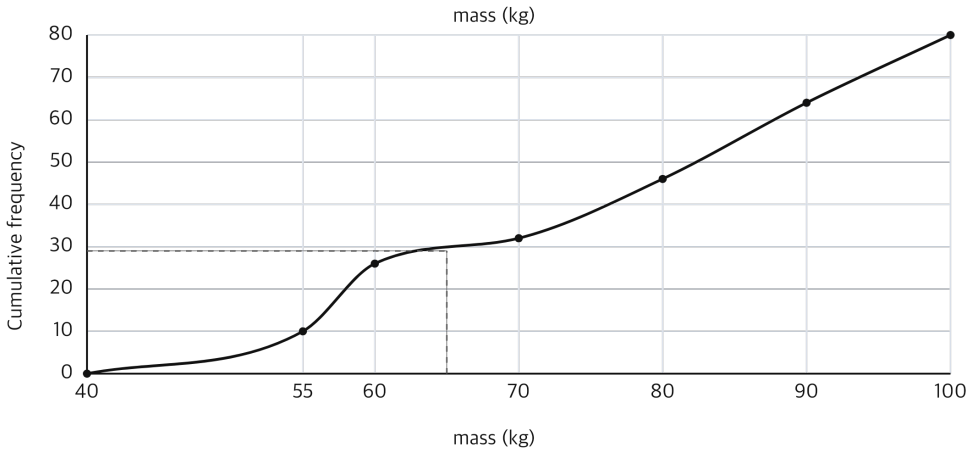
37 plants

Teacher answers & working

Cumulative Frequency Curves and Ogives · Reading Curves

STA-5424-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Estimate the number of parcels with mass above 65 kg.

Answer:

51 parcels

Working:

Read the cumulative frequency at 65 kg.

number up to 65 ≈ 29

For the number above this value, subtract from the total frequency.

80 - 29 = 51

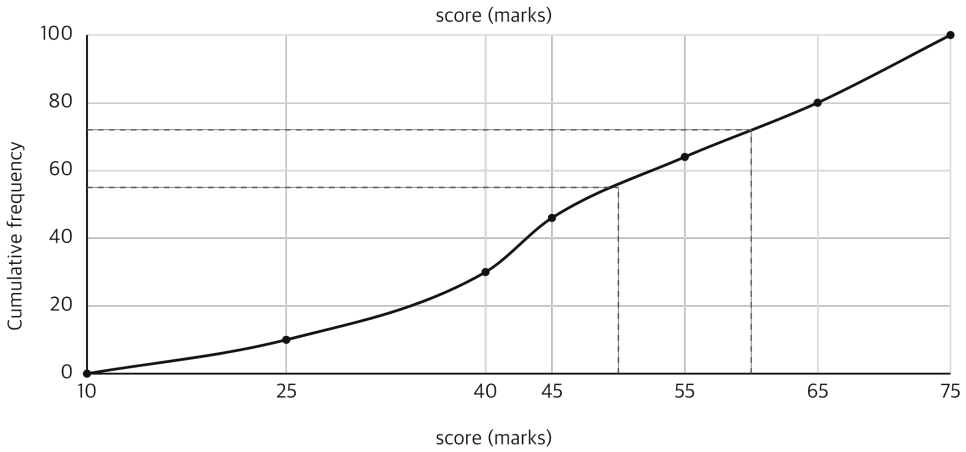
51 parcels

Teacher answers & working

Cumulative Frequency Curves and Ogives · Reading Curves

STA-5424-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Estimate the number of candidates with score between 50 and 60 marks.

Answer:

17 candidates

Working:

Read the cumulative frequency at 50 marks and 60 marks.

CF at $50 \approx 55$

CF at $60 \approx 72$

Subtract the lower cumulative frequency from the higher cumulative frequency.

$72 - 55 = 17$

17 candidates